

What exactly is weakly informative prior?

Asked 11 years, 1 month ago Modified 11 years, 1 month ago Viewed 9k times



Is there a precise definition of weakly informative prior?

11

How is it different from a subjective prior with broad support?



bayesian prior



Share Cite Improve this question Follow

asked Oct 22, 2013 at 16:22



Memming

1,662 13 23

- 3 My understanding is that a weakly-informative prior expresses more about the researcher's attitude towards the prior, rather than any mathematical properties of the prior itself. The canonical example would be Gelman's recommendation of a Cauchy prior with location 0 and scale 5/2 for logistic regression. – Sycorax ♦ Oct 22, 2013 at 16:24

The basic idea is to strike a balance between the least amount of prior information possible while also regularizing a potentially ill-behaved likelihood and thereby gaining computational efficiency. An example might be a prior that rules out negative variances, or the [LKJ prior regularizing the correlation matrix to have unit diagonals](#). – Durden Jun 21, 2023 at 4:45

2 Answers

Sorted by: Highest score (default)



The above comment is accurate. For a quantitative discussion, there are a number of "uninformative" priors in the literature. See for example Jeffreys' prior; see earlier post [What is an "uninformative prior"? Can we ever have one with truly no information?](#)

8

They are defined in different ways, but the key is that they do not place too much probability in any particular interval (and hence favor those values) with the uniform distribution being a canonical example. The idea is to let the data determine where the mode is.



Share Cite Improve this answer Follow

edited Apr 13, 2017 at 12:44

answered Oct 22, 2013 at 16:48



1

user31668

So weakly informative prior is just a better name for slightly informative "uninformative prior"? – Memming Oct 22, 2013 at 17:09

- 2 I usually use "uninformative" as that is more common and indicates intent. Weakly informative is probably more accurate though, as all distributions carry some information (**unless** they are improper priors..but that is another discussion) – user31668 Oct 22, 2013 at 17:13

- 1 I had the impression that *weakly informative* priors aim to avoid having to commit to *uninformative* priors formally defined according to some theory or other - they're proper priors that work for inference, while not taking into account all prior knowledge as a fully subjective prior would. – Scortchi ♦ Oct 23, 2013 at 10:49

- 1 @Scortchi: I think your comment highlights the ambiguity inherent in "weakly informative prior." Your interpretation makes sense and is in a similar vein to user777. The relationship between "informativeness" and probability is a tricky thing, with only partially satisfactory solutions (e.g., Shannon entropy). I get your point though...they are not necessarily synonymous, as weakly informative priors may use only some of the information, while uninformative priors explicitly ignore all available information. – user31668 Oct 23, 2013 at 12:07



Further to Eupraxis1981's discussion of informative priors, you can think of the "information" in a prior as inversely proportional to its variance. Consider a prior with near zero variance: you're basically saying "before looking at the data, I'm almost positive I already know the location of the true value of the statistic." Conversely, if you set a really wide variance, you're saying "without looking at the data, I have really no assumptions about the true value of the parameter. It could be pretty much anywhere, and I won't be that surprised. I've got hunch it's probably near the mode of my prior, but if it turns out to be far from the mode I won't actually be surprised."



Uninformative priors are attempts to bring no prior assumptions into your analysis (how successful they are is open to debate). But it's entirely possible and sometimes useful for a prior to be only "weakly" informative.



Share Cite Improve this answer Follow

edited Oct 22, 2013 at 17:02

answered Oct 22, 2013 at 16:59



Sycorax ♦

93.8k 23 235 390



David Marx

7,137 2 30 46